In-Class Exercise

ECON 300 – Labour Economics

Winter 2026

Time: 30–40 minutes
Format: Small groups (3–4 students)
Deliverable: Annotated labour–leisure diagrams and short written answers

Learning Objectives

By the end of this exercise, students should be able to:

- Explain how different **demogrant designs** affect labour supply using the labour–leisure model.
- Distinguish **income effects** from **substitution effects**.
- Predict effects on:
  - hours worked (intensive margin), and
  - labour force participation (extensive margin).
- Understand why policy **design**, not just generosity, matters for work incentives.

Common Setup (Applies to All Groups)

Each individual has:

- Total available time: $T = 80$ hours per week
- Hours worked: $h = 80 - L$
- Market wage: $w = 20$ dollars per hour

Baseline (no demogrant) budget constraint:

$$C = 20h$$

Assume:

- Leisure is a **normal good**
- Individuals can choose $h = 0$ (non-participation)

Policy Experiments: Types of Demogrants

Each group is assigned **one** of the following demogrant designs.
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Policy A: Universal Lump-Sum Demogrant

Every adult receives \$400 per week, regardless of hours worked.

$$C = 400 + 20h$$

Tasks

1. Draw the new budget constraint on a labour–leisure graph.
2. Compare it to the baseline budget line.
3. Identify:
   - the income effect, and
   - whether there is a substitution effect.
4. Predict the effect on:
   - hours worked, and
   - labour force participation.

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Policy B: Targeted Demogrant (Non-Workers Only)

Individuals receive \$400 **only if they do not work**.

$$C = \begin{cases} 400 & \text{if } h = 0, \\ 20h & \text{if } h > 0. \end{cases}$$

Tasks

1. Draw the full budget set, carefully noting any kinks or discontinuities.
2. Explain how this policy affects the **reservation wage**.
3. Which margin is most affected:
   - labour force participation, or
   - hours worked among participants?
4. Predict whether more or fewer people choose to work at all.

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Policy C: Phased-Out Demogrant (Negative Income Tax Style)

Individuals receive \$400, but benefits are reduced by \$0.50 for each dollar earned.

$$C = 400 + (1 - 0.5) \cdot 20h = 400 + 10h$$

Tasks

1. Compare the slope of this budget constraint to the baseline.
2. Identify:
   - the income effect, and
   - the substitution effect.
3. Predict the effect on:
   - hours worked conditional on working, and
   - the likelihood of labour force participation.

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Policy D: Conditional Demogrant (Work Requirement)

Individuals receive \$400 only if they work **at least 20 hours per week**.

$$C = \begin{cases} 20h & \text{if } h < 20, \\ 400 + 20h & \text{if } h \geq 20. \end{cases}$$

Tasks

1. Draw the budget constraint and clearly mark the eligibility threshold.
2. Explain how this policy affects individuals who would otherwise work close to 20 hours.
3. What type of behavioural distortion does this policy create?
4. Predict whether hours worked may cluster around specific values.

Wrap-Up Discussion (All Groups)

Answer briefly:

1. Which demogrant design creates:
   - only an income effect?
   - both income and substitution effects?
2. Which policy is most likely to:
   - reduce labour force participation?
   - reduce hours worked among those who still work?
3. Why might governments prefer **conditional** or **phased-out** demogrants over universal ones?

Key Takeaways

- All demogrants generate income effects.
- Only some designs distort the price of leisure.
- Participation responses often dominate hours responses.
- Policy design matters as much as policy generosity.